Measurement Scales, Likert Parameters, and Statistical Decision-Making in Senior High School Quantitative Research

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ABSTRACT

Senior high school researchers often encounter difficulty not in collecting data but in deciding how the data should be classified, summarized, and interpreted. This article reorganizes the introductory statistical concepts presented in Statistics Made Easy into a practical decision framework linking research questions, measurement scales, descriptive summaries, Likert-type response parameters, and hypothesis direction. The discussion distinguishes nominal, ordinal, interval, and ratio data as presented in the source text; explains how the source constructs verbal-interpretation intervals for a five-point Likert scale; and connects one-tailed and two-tailed testing to the wording of an alternative hypothesis. The article emphasizes a sequence of decisions: identify the research question, determine the expected data type, select a compatible summary or test, define interpretation rules before analysis, and report the statistical decision transparently. It also identifies points in the source that require editorial verification before publication, particularly the coexistence of weighted-mean practice with a later recommendation to summarize Likert responses using the median or mode. Rather than resolving that tension without evidence, the article treats it as a methodological choice that must be explicitly justified. The resulting framework is intended as a classroom-ready guide for student-researchers and teachers who need a structured path from questionnaire design to data interpretation.

Keywords*:* *measurement scales; Likert scale; statistical analysis; hypothesis testing; senior high school research; quantitative research*

INTRODUCTION

Quantitative research requires more than the mechanical use of formulas. Before any computation is performed, the researcher must know what kind of information the research question is expected to produce. Statistics Made Easy presents this problem as one of alignment: the research question influences the research methodology, research design, and statistical tool, while the nature of the data limits which summaries and tests are meaningful.

For beginning researchers, this alignment is easy to overlook because spreadsheet and statistical software can calculate a large number of outputs even when the selected procedure is not well matched to the data. A practical teaching strategy is therefore to place classification and decision-making before computation. The purpose of this article is to reorganize the source book's introductory material into a compact framework that can be used while planning a senior high school quantitative study.

From Research Question to Data Type

The source text describes statistics as the gathering, analysis, interpretation, and presentation of empirical data. It differentiates descriptive statistical tools, which summarize observed data, from inferential procedures, which are used when the researcher makes judgments about a population from sample information. This distinction becomes useful only after the researcher has identified the type of variable being studied.

The book presents four measurement scales. Nominal data are categorical and are illustrated through respondent characteristics such as gender, educational attainment, and other socioeconomic categories.

Ordinal data express rank or order. Interval data are described through equally spaced response categories, with the Likert scale given as the central example. Ratio data are presented as ordered numerical measurements that also possess a true-zero property.

A useful classroom habit is to write the expected data type beside every research question before preparing a questionnaire or spreadsheet. This forces the researcher to think about whether the answer will be a category, rank, rating, or numerical measurement and helps prevent the statistical tool from being chosen only after the data have already been collected.

A Practical Measurement-to-Analysis Matrix

Scale Primary characteristic in the source Typical classroom use Common summary emphasized in the book

Frequency, percentage, mode

Nominal Categories without an ordered scale
Profiles such as educational attainment or occupation

Median or mode

Ordinal Responses can be ranked or ordered
Ranked preferences or ordered categories
Likert-type rating examples in the source
Interval Ordered values with equal intervals as presented in the book
Mean/weighted mean in later worked examples; source also discusses median/mode Ratio Ordered numerical values with a true zero
Quantitative measurements with an absolute zero
Mean and other numerical summaries

Constructing Likert Parameters Before Analysis

The book treats the Likert scale as a common response format for questionnaire-based quantitative research. For a five-point scale, it demonstrates a parameter-building procedure that begins with the numerical endpoints, obtains an interval width, and then builds interpretation bands from the lowest to the highest response. In the source example, the five-point range is divided into equal-width bands and paired with verbal descriptions such as Always, Often, Sometimes, Rarely, and Never.

The practical value of establishing these parameters before analysis is consistency. If the researcher determines the verbal interpretation only after seeing the results, the thresholds can be unconsciously adjusted to favor a preferred conclusion. A better workflow is to define the scale, coding, verbal labels, and interpretation bands in the methodology section before the data are summarized.

The same principle applies when four-point or other rating scales are used. The numeric coding and verbal interpretation should be declared before analysis, and the same rule should be applied to all indicators within the same scale unless the research design provides a documented reason for a different treatment.

Descriptive Summaries and a Source-Level Methodological Tension

Statistics Made Easy discusses mean, median, mode, frequency, percentage, and weighted mean as commonly used descriptive tools. It also demonstrates weighted-mean practice for Likert-type questionnaire data in later chapters. However, the introductory chapter separately states that Likert-scale data may be summarized using the median or mode rather than the mean, with the mode described as a straightforward option for interpretation.

This draft does not silently reconcile those two positions. For publication, the article recommends that the researcher clearly state the convention being followed and explain why it is appropriate for the study's instrument and reporting purpose. The important editorial principle is transparency: the manuscript should not present one treatment as universally required when the source itself contains more than one instructional approach.

Directional and Non-Directional Hypotheses

The source connects hypothesis wording to the selection of a one-tailed or two-tailed test. A one-tailed test is described as directional: the alternative hypothesis specifies that a value is greater, higher, lower, smaller, increased, or decreased. A two-tailed test is described as non-directional: the researcher asks whether a difference exists without specifying the direction in advance.

For student researchers, the main lesson is that test direction should be decided from the hypothesis before examining the results. The wording of the alternative hypothesis should not be changed after the data are analyzed merely to make a result appear significant. A short hypothesis-planning worksheet can therefore contain four fields: research question, null hypothesis, alternative hypothesis, and intended tail of the test.

Proposed Five-Step Statistical Decision Framework

Step 1 - Define the research question. Determine whether the study seeks a description, rank, relationship, difference, or other statistical conclusion.

Step 2 - Identify the expected data type. Classify the principal variables using the measurement scheme adopted in the study.

Step 3 - Predefine the coding and interpretation rule. For rating scales, document the numerical codes and verbal interpretation bands before analysis.

Step 4 - Match the statistical procedure to the question and data. Use descriptive summaries for description and inferential tools when the research question asks about relationships or differences.

Step 5 - Report the decision with the supporting statistic. Present the relevant computed value, significance level or comparison rule, and an interpretation that answers the research question without overstating what the procedure establishes.

Implications for Teaching Quantitative Research

The source book was developed to make statistical analysis more accessible to senior high school researchers, particularly those who experience anxiety during data analysis and interpretation. A decision- first approach supports that purpose because it reduces statistics to a sequence of documented choices rather than a collection of disconnected formulas.

Teachers may use the framework as a pre-analysis checklist. Before students are allowed to compute, they should be able to identify the variable type, justify the statistical tool, state the interpretation rule, and explain whether the hypothesis is directional or non-directional. This shifts classroom attention from software operation to statistical reasoning.

CONCLUSION

The appropriate use of statistics begins before a spreadsheet is opened. Measurement scale, research question, hypothesis direction, response coding, and interpretation rules should be aligned in advance. By reorganizing the introductory material of Statistics Made Easy into a decision framework, this article provides a practical structure for senior high school researchers who need to move from questionnaire design to defensible data interpretation. The framework does not replace statistical judgment; rather, it makes the decisions visible and therefore easier to teach, review, and improve.

REFERENCE

Simpal, M. T., & Mangakoy, A. B. (2022). Statistics made easy: Commonly used statistical tools and its computer applications in quantitative research for senior high school (1st ed.). Shirley Publishing House.